

Flyhomes BBYS Calculator

Calculation Logic — Engineering Handoff Spec

Source of truth: `index.html` (single-file prototype), as implemented on 2026-08-14 · Verified against live JS, not just design docs · Supersedes the 2026-08-13 revision

Purpose. This document specifies the calculation and decision logic exactly as implemented in the working prototype (`index.html`), so it can be re-implemented in production code. Where the prototype's behavior differs from — or isn't covered by — the design doc (`CLAUDE.md`), that is called out explicitly with a **GAP** or **BUG** tag. Items resolved since the last revision are tagged **RESOLVED** . Everything else is marked **MATCHES DOC** .

Status as of this revision: a code-vs-design-doc review by engineering (against archived Notion docs, not this spec) surfaced three additional issues in the Cash Offer / Total Cost formulas, all now resolved — see §6 for the full change log. No open gaps or bugs as of 2026-08-14.

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1. State model

A single mutable state object `S` drives every calculation. Relevant fields ([index.html:1150–1178](#)):

Field	Meaning	Default
<code>homeValue</code>	Departing home value (current home) — also the GBC fee-tier input, capped at \$2,000,000 (§1.1)	\$1,200,000
<code>mortgage</code>	Current mortgage balance on departing home	\$300,000
<code>newPrice</code>	Estimated purchase price of new home	\$800,000
<code>assets</code>	Liquid cash assets available — the <i>sole</i> basis for Cash Offer eligibility (§3.2, §6.7)	\$40,000
<code>transitionDays</code>	Days between buying new home and selling old one. User-editable, min 1	60
<code>gbc</code> , <code>cash</code> , <code>noCash</code> , <code>downsize</code>	Resolved product flags for the current render (derived — see §2)	—
<code>cashIntent</code> , <code>gbc_base</code> , <code>noCash_base</code>	Raw user intent flags, before Rule 3/5 gating resolves them	—
<code>forceCashOffer</code>	"Option A" manual override: user chose standalone Cash Offer over an auto-recommended Cross Collateral. Reset to <code>false</code> whenever challenge selection changes	false
<code>naturalIsCC</code>	True if Cross Collateral would be recommended <i>without</i> the override — drives the switch-offer banner	—
<code>statesDiffer</code>	True when "current home state" and "buying state" dropdowns are both filled and unequal	false
<code>flyhomesLoan</code>	True = GBC bundle rate; false = GBC standalone rate	true
<code>savCashPct</code> / <code>savStagedPct</code> / <code>savHpaPct</code>	Savings-section sliders	3.5% / 5% / 4%
<code>savStagingCost</code>	Flat staging cost deducted from staged-home premium	\$3,500
<code>savMovingFee</code> , <code>savHousing</code> , <code>savStorage</code>	"Move Once" line items — auto-computed, user-editable	derived from <code>homeValue</code>

`availableEquity` / `combinedResources` — present in earlier revisions of this spec — no longer exist in the code. Cash Offer now uses `assets` directly; see §3.2 and §6.7.

1.1 Home-value input ceiling

"Estimated Value of Your Current Home" is capped at **\$2,000,000** — both the slider max ([index.html:325](#)) and the effective usable range of the paired free-text field, enforced in `render()` ([index.html:1544–1552](#)). Above that value,

`#results-panel` is hidden entirely and a "Please contact us to learn more." message is shown instead — no estimate is attempted. This keeps the GBC >\$2M exception path (§4.3) unreachable through the UI. See §6.5 for why.

2. Challenge selection → product rule engine

Four challenge checkboxes map to internal ids: `dti`, `equity`, `cash`, `downsize` ([index.html:1654–1660](#)), stored in a Set called `selected`. Two functions resolve them into product flags on every change: `syncStateFromChallenges()` then `applyRule3Gating()`. `applyRule3Gating()` also re-runs on every `render()` so slider changes (price, mortgage, home value) can flip the recommendation live.

2.1 `syncStateFromChallenges()` [index.html:1662–1676](#)

```
isDownsize      = selected.has('downsize')
isCrossCollateral = (selected.has('equity') && selected.has('cash')) || isDownsize

S.downsize      = isDownsize
S.noCash_base   = isCrossCollateral
S.cashIntent    = selected.has('cash') && !isCrossCollateral
S.gbc_base      = selected.has('dti') && !isCrossCollateral
S.forceCashOffer = false // reset override on any challenge-card change
```

Matches design doc Rule 0 (downsize overrides) and Rule 1 (equity+cash → Cross Collateral).

MATCHES DOC

2.2 `applyRule3Gating()` [index.html:1726–1779](#)

```
isRule3CC = false
if (cashIntent && newPrice > 0):
    projectedFinalLTV = (newPrice + mortgage - homeValue * 0.9) / newPrice
    isRule3CC = projectedFinalLTV <= 0.75

// Cash-shortfall fallback to Cross Collateral ($6.8) – only reached when the LTV
// check above did NOT already route to CC
if (!isRule3CC):
    cashEligible = assets >= newPrice * 0.05
    if (!cashEligible):
        ccCltv = (newPrice + homeValue) * 0.75 - mortgage
        ccMaxLoan = max(0, min(newPrice * 1.05, ccCltv))
        if (ccMaxLoan + assets >= newPrice): isRule3CC = true

if (statesDiffer): // Rule 5 – cross-state block
    if (noCash_base): // a same-state CC would have won
        if (!downsize): cashIntent = true // Rule 1 combo → keep the cash-offer intent
        gbc_base = true // both fallbacks pair with DTI Buster
    noCash_base = false
    isRule3CC = false

naturalIsCC = noCash_base || isRule3CC

if (naturalIsCC && !downsize && forceCashOffer):
    // Option A: user overrode CC → standalone Cash Offer
    noCash = false; cash = true; gbc = gbc_base
else:
    noCash = naturalIsCC
```

```
cash      = cashIntent && !isRule3CC
gbc       = isRule3CC ? false : gbc_base    // CC can't pair with GBC
```

Matches design doc Rule 3 (auto-gate on `projectedFinalLTV`, with the "Keep More Cash" override), Rule 5 (cross-state block), and the new cash-shortfall fallback (§6.8). **MATCHES DOC**

NEW

since 2026-08-13: cash-shortfall fallback to Cross Collateral. When Rule 3's LTV check would recommend Cash Offer but the buyer's liquid cash can't cover the 5% down payment, check whether Cross Collateral's own loan plus the buyer's cash would cover the purchase price (`ccMaxLoan + assets >= newPrice`) — *not* a fixed percentage threshold, since Cross Collateral has no minimum-down-payment rule of its own. If it covers the gap, switch to Cross Collateral (eliminates the cash requirement entirely). If Cross Collateral would also leave an uncovered gap, keep recommending Cash Offer with the existing cash-shortfall warning. Full write-up in §6.8.

2.3 Card visibility & labels — `applyProductLabel()` [index.html:1781–1815](#)

Condition	Result
<code>dtiOnly = selected.size===1 && has('dti')</code>	Hides main solution card, cost/benefit card, CTA card, number card. Shows only the GBC/DTI Buster card(s).
<code>noneSelected = selected.size===0</code>	Same as above — nothing shown.
<code>showGBC = !noneSelected && !isCrossCollat</code>	GBC (DTI Buster) card shows whenever anything is selected and the resolved product isn't Cross Collateral — regardless of whether "dti" was checked. This is what implements Rule 2/Rule 3's "+ DTI Buster" pairing.
<code>mostLikely = showGBC && !has('dti')</code>	"Recommended for you" badge on the DTI Buster card — shown when GBC is auto-added by Rule 2/3/5 rather than manually checked.

DTI-only mode renders two separate cards, not one merged card: a standalone "DTI Buster" title card (`#gbc-solution-card`) and a separate "Estimated Cost + Estimated Total Value" card (`#gbc-dti-costs`), toggled via a `.dti-solo` class on `#solution-card-wrapper` . This only applies in DTI-only mode — the combined "DTI Buster + another product" scenario still renders as one merged card with an internal divider, unchanged.

3. Core product formulas

Computed in `calc()` <index.html:1229-1294>, branching on the resolved `S.noCash` / `S.cash` flags from §2:

```
cashThreshold = newPrice * 0.05
cashEligible  = assets >= cashThreshold // liquid cash only – see §6.7
gbcActive     = S.gbc && !noCash        // GBC never pairs with Cross Collateral
```

3.1 Branch A — Cross Collateral (if `noCash` is true)

```
ccCltv  = (newPrice + homeValue) * 0.75 - mortgage
maxLoan  = max(0, min(newPrice * 1.05, ccCltv))
ltvAcq   = newPrice > 0 ? maxLoan / newPrice : 0
originationRate = ltvAcq > 0.90 ? 0.015 : 0.010
brokerFee      = maxLoan * 0.01

// Label:
if (S.downsize): solutionName = "Retire & Downsize"      productName = "Cross Collateral"
else:            solutionName = "Move with $0 out of pocket" productName = "Cross Collateral"
```

Matches design doc §5E Cross Collateral formula and §5B origination-fee tier.

MATCHES DOC

3.2 Branch B — Cash Offer (`else if cash && cashEligible`)

```
maxLoan  = max(0, newPrice * 0.95)
ltv       = newPrice > 0 ? maxLoan / newPrice : 0
originationRate = ltv > 0.90 ? 0.015 : 0.010
brokerFee      = maxLoan * 0.01
solutionName = "All-Cash Advantage"  productName = "Cash Offer"
```

Matches design doc §5E "BBYS + Cash Offer" formula.

MATCHES DOC

RESOLVED since 2026-08-13, two changes:

- 1. Eligibility gate fixed (§6.7):** `cashEligible` now checks `assets` alone. The previous version added unlockable home equity (`combinedResources = assets + availableEquity`) — a bug, since that equity isn't actually accessible for a Cash Offer down payment unless the buyer is also using Instant Equity or Cross Collateral.
- 2. Loan formula simplified (§6.9):** the previous formula was `min(newPrice, assets / 0.05) * 0.95`. Once `cashEligible` already requires `assets >= newPrice * 0.05` to even enter this branch, `assets / 0.05 >= newPrice` is guaranteed, so the `min()` always resolved to `newPrice` — it never actually capped anything. Removed as dead logic; output is unchanged for every input.

3.3 Branch C — Instant Equity (else — default / fallback)

```
maxLoan  = max(0, homeValue * 0.75 * 0.9 - mortgage)
originationRate = 0.02 // 1st lien rate, applied unconditionally – see §6.2
```

```
brokerFee = maxLoan * 0.005
```

```
solutionName = "Equity for Down Payment"    productName = "Instant Equity"
```

Matches design doc §5E Instant Equity formula and §5B (1st-lien rate). **MATCHES DOC**

This branch is also the fallback whenever `cash` is true but `cashEligible` is false (insufficient liquid cash for the 5% down payment, and Cross Collateral's own loan didn't cover the gap either — §2.2) — in that case a "cash warning" banner is additionally shown ([index.html:1652–1658](#)) with the shortfall amount: `gap = cashThreshold - S.assets`, text "You need \$X more in cash."

4. Fee formulas

4.1 Origination fee

```
originationFee = maxLoan * originationRate // rate resolved per branch above, §3
```

4.2 Broker fee

Rolled into "One Time Fees" for display (previously labeled "Other Costs") — never shown as a standalone line item to the buyer, per design doc §5D. Rate depends on product track (see §3): Cross Collateral 1% of capped loan, Cash Offer 1% of loan amount, Instant Equity 0.5% of advance amount.

4.3 GBC fee — `getGbcFee(homeValue, isBundle)` [index.html:1215–1226](#)

```
if      (homeValue <= 500,000):  standalone = 2,500;  credit = 0
elif    (homeValue <= 750,000):  standalone = 3,500;  credit = 1,000
elif    (homeValue <= 1,000,000): standalone = 5,000;  credit = 2,500
elif    (homeValue <= 1,500,000): standalone = 7,500;  credit = 2,500
elif    (homeValue <= 2,000,000): standalone = 10,000; credit = 2,500
else:                                     return null // unreachable – see §1.1

return isBundle ? (standalone - credit) : standalone
```

MATCHES DOC §5C exactly. The `else: return null` branch is defensive dead code — the §1.1 input ceiling means `homeValue` can never exceed \$2,000,000 through the UI.

4.4 Accrued interest

```
accruedInterest = maxLoan * 0.0999 * (transitionDays / 365)
```

Only shown/rendered when `maxLoan > 0` (hidden for DTI-only). **MATCHES DOC**

4.5 Total estimated cost

```
gbcFeeNum = gbcFee ?? 0 // getGbcFee's null branch is unreachable per §1.1/§4.3
total      = originationFee + brokerFee + gbcFeeNum + accruedInterest
```

MATCHES DOC §5A exactly: `Origination Fee + Broker Fee + GBC Fee + Accrued Interest` — all four terms. See §6.6 for a documentation-only fix (the doc's headline formula previously omitted Broker Fee, even though §5D always required it to be included, just bundled into "One Time Fees" rather than broken out).

5. Estimated Savings formulas

Computed in `calcSavings()` <index.html:1366-1378>:

5.1 Cash Offer Price Advantage

```
isCash      = cash || noCash
cashDiscount = isCash ? newPrice * (savCashPct / 100) : 0
```

Matches design doc §5F.1: shows whenever the recommended product uses a cash offer — Cash Offer track (`S.cash`) or Cross Collateral / Retire & Downsize (`S.noCash`). **MATCHES DOC**

5.2 Staged Home Premium

```
staged = max(0, homeValue * (savStagedPct / 100) - savStagingCost)
```

MATCHES DOC (tooltip version, net of \$3,500 staging cost)

5.3 Home Price Appreciation (HPA)

```
hpa = newPrice * (savHpaPct / 100) * (transitionDays / 30) / 12
```

Buy-side only, per doc. **MATCHES DOC**

5.4 "Move Once, Not Twice" (logistics)

```
savMovingFee = round(homeValue * 0.003 + 500)
savHousing   = round(homeValue * 0.006 * (transitionDays / 30))
savStorage   = round((150 + homeValue * 0.0002) * (transitionDays / 30))
logistics    = savMovingFee + savHousing + savStorage
```

MATCHES DOC exactly.

5.5 Grand total

```
total = cashDiscount + staged + hpa + logistics
```

The DTI-only card computes its own total that always excludes the cash discount row: `g-total = staged + hpa + logistics` . It also shows its own "Every \$1 you spend..." ROI line, using the GBC fee (not the shadow Instant Equity total) as the cost basis — see §6.4.

6. Change log: issues found & resolved

No open gaps or bugs as of this revision. Entries 6.1–6.6 were resolved by 2026-08-13; entries 6.7–6.9 were raised by an engineering code-vs-design-doc review (against archived Notion docs) on 2026-08-14 and resolved the same day.

RESOLVED 6.1 Downsize + cross-state mismatched label vs. math

Fixed by deriving the solution card's label from the same resolved `calc()` branch that computes the numbers, instead of a separate, duplicate label-derivation path.

CONFIRMED, NOT IMPLEMENTED 6.2 Instant Equity 2nd-lien rate

Confirmed intentional: this is a directional estimate tool, and lien position can't be reliably inferred from any existing input. `originationRate` stays hardcoded at 2.0% for all Instant Equity scenarios.

RESOLVED 6.3 Cash-discount eligibility differed between main card and savings section

Same fix as §6.7 below — both now key off the same liquid-cash check.

RESOLVED 6.4 Dead code: unused savings field in `calc()`; ROI tooltip never rendered

Removed the unused `savings` field. Separately, the "Every \$1 you spend..." tooltip on the main card never rendered on hover — its `tip-box` was a `<div>` nested inside a `<p>`, which is invalid HTML; the browser silently split the paragraph and ejected the div outside its `.tip` wrapper, breaking the CSS `:hover` relationship. Fixed by using a `<span>` instead.

RESOLVED (differently) 6.5 GBC fee >\$2M exception silently excluded from grand total

Resolved upstream instead of downstream: rather than teaching the total-cost calculation to flag an incomplete state, the home-value input itself is now capped at \$2,000,000 (§1.1) — the exception path is unreachable.

RESOLVED 6.6 Total Estimated Cost formula omitted Broker Fee in the doc's headline formula

Raised by engineering: two archived Notion doc variants disagreed with each other (one omitted Broker Fee, the other omitted Accrued Interest) and both were incomplete relative to the prototype. CLAUDE.md §5A now states all four terms explicitly: `Origination Fee + Broker Fee + GBC Fee + Accrued Interest`. No code change — the prototype's `total` calculation (§4.5) was already correct.

RESOLVED 6.7 Cash Offer eligibility incorrectly counted home equity

Raised during a Cash Offer Max Loan formula review: the prototype's `combinedResources = assets + availableEquity` let a buyer with little liquid cash but high home equity qualify for Cash Offer, despite having no way to actually access that equity for a down payment (it's still locked in the unsold departing home unless the buyer is also using Instant Equity or Cross Collateral). Fixed: `cashEligible` and the Cash Offer max loan calculation now use `assets` alone. `availableEquity / combinedResources` were removed from `calc()` entirely — see §3, §3.2.

RESOLVED 6.8 New rule: cash-shortfall fallback to Cross Collateral

When Cash Offer would otherwise be recommended (Rule 3, `projectedFinalLTV > 0.75`) but the buyer's liquid cash can't cover the 5% down payment, `applyRule3Gating()` now checks whether Cross Collateral's own loan, combined with the buyer's cash, would cover the full purchase price (`ccMaxLoan + assets >= newPrice`) — not a fixed percentage of price, since Cross Collateral carries no minimum-down-payment rule of its own. An earlier version of this check used `ccMaxLoan >= newPrice * 0.95` (matching Cash Offer's own qualifying bar) — that undercounts real shortfalls whenever the CC loan clears 95% but still leaves a gap larger than the buyer's cash. If Cross Collateral would also leave a gap, Cash Offer stays recommended with the existing cash-shortfall warning. See §2.2.

RESOLVED**6.9 Cash Offer max loan formula simplified (no behavior change)**

The prototype's original `min(newPrice, assets / 0.05) * 0.95` is mathematically equivalent to the simpler `newPrice * 0.95` , given that `cashEligible` already gates entry into this branch on `assets >= newPrice * 0.05` — which guarantees `assets / 0.05 >= newPrice` is always true by the time the loan amount is computed. The `min()` never actually capped anything; removed as dead logic. Computed output is identical for every input — see §3.2.

7. Appendix: constants reference

Constant	Value	Used in
Accrued interest rate	9.99% / yr	Total cost — all tracks
Instant Equity multiplier	$0.75 \times 0.9 = 0.675$	Instant Equity value
Cash Offer max loan-to-value	95%	Cash Offer max loan cap
Cash Offer eligibility threshold	Liquid Assets $\geq$ 5% of purchase price	Cash Offer branch gate (liquid cash only, §6.7)
Cross Collateral acquiring LTV cap	105% of purchase price	Cross Collateral max loan cap
Cross Collateral combined LTV cap	75% of (purchase price + current home value)	Cross Collateral CLTV formula
Origination fee — CO / CC, LTV > 90%	1.5%	Origination fee
Origination fee — CO / CC, LTV $\leq$ 90%	1.0%	Origination fee
Origination fee — Instant Equity	2.0% (all scenarios)	2nd-lien rate intentionally not implemented, §6.2
Broker fee — Cross Collateral / Cash Offer	1% of loan amount	Bundled into One Time Fees
Broker fee — Instant Equity	0.5% of advance amount	Bundled into One Time Fees
Rule 3 gating threshold	$\text{projectedFinalLTV} \leq 0.75 \rightarrow$ Cross Collateral	Auto-recommendation between CC and CO
Cash-shortfall $\rightarrow$ CC fallback	CC Max Loan + Liquid Assets $\geq$ New Home Price	Rule 3 fallback (§2.2, §6.8)
Home-value input ceiling	\$2,000,000	Blocks results panel + GBC exception path (§1.1)
Staging cost (flat, editable)	\$3,500	Staged Home Premium
Default sliders	Cash discount 3.5%, Staged premium 5%, HPA 4%	Estimated Savings
Default transition days	60 (min 1)	Accrued interest, HPA, temp housing, storage